VLADIMIR, Russian Federation

WMO: **275320**

Lat: **56.117N** Lon: **40.350E** Elev: **172** StdP: **99.28** Time Zone: **3.00 (E03)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-25.2	-22.0	-28.1	0.3	-24.9	-24.7	0.4	-21.8	9.4	-4.8	8.4	-4.2	2.3	330	0.583	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	9.3	29.7	20.9	27.7	20.1	25.9	19.3	22.3	27.7	21.0	26.3	19.9	24.7	2.9	150	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	20.5	15.4	25.8	19.1	14.2	24.1	18.1	13.3	22.9	66.4	27.5	61.6	26.3	57.9	24.8	27.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	8.4	7.5	6.6	DB	-28.1	31.8	3.6	2.5	-30.7	33.6	-32.8	35.0	-34.8	36.4	-37.4	38.2	(4)
(5)				WB	-28.3	23.9	3.6	2.0	-30.8	25.4	-32.9	26.6	-34.9	27.7	-37.5	29.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		5.2	-8.7	-8.0	-2.5	6.2	12.9	16.6	19.4	17.1	11.5	5.0	-1.8	-6.1	(6)
(7)		DBStd		11.17	6.92	6.90	4.84	5.16	5.09	4.12	3.71	3.86	4.04	4.53	5.34	6.47	(7)
(8)		HDD10.0		2686	581	503	387	137	28	2	1	1	28	164	355	498	(8)
(9)		HDD18.3		4930	839	737	645	364	180	81	32	70	207	413	605	757	(9)
(10)		CDD10.0		937	0	0	0	23	118	200	292	220	74	10	0	0	(10)
(11)		CDD18.3		138	0	0	0	0	11	29	64	31	2	0	0	0	(11)
(12)		CDH23.3		1110	0	0	0	2	99	241	510	247	11	0	0	0	(12)
(13)		CDH26.7		293	0	0	0	0	21	59	143	71	0	0	0	0	(13)
(14)	Wind	WSAvg		3.2	3.4	3.4	3.5	3.2	3.3	3.0	2.5	2.6	2.8	3.4	3.4	3.5	(14)
(15)	Precipitation	PrecAvg		616	44	33	33	37	47	81	65	61	57	63	49	46	(15)
(16)		PrecMax		801	92	74	70	78	96	174	161	129	187	181	114	92	(16)
(17)		PrecMin		469	20	9	4	14	10	7	4	18	12	18	5	17	(17)
(18)		PrecStd		89	17	13	16	16	22	41	38	33	35	40	25	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		3.2	3.4	11.8	23.2	29.0	30.8	32.7	31.9	24.9	18.6	10.6	6.4	(19)
(20)			MCWB		2.2	1.4	7.1	14.4	19.3	20.2	22.6	23.3	18.2	14.4	9.1	5.6	(20)
(21)		2%	DB		1.6	2.0	7.7	20.0	26.0	28.3	29.9	28.4	22.3	15.2	7.6	4.1	(21)
(22)			MCWB		0.8	0.7	4.0	12.6	18.2	19.8	21.1	21.2	16.6	11.9	6.7	3.3	(22)
(23)		5%	DB		0.8	1.2	5.2	17.2	23.4	26.2	28.0	25.8	19.9	13.0	6.0	2.4	(23)
(24)			MCWB		0.2	0.3	2.5	11.5	16.6	19.0	20.6	19.6	15.3	10.9	5.1	1.6	(24)
(25)		10%	DB		0.0	0.5	3.4	14.8	21.3	23.9	26.2	23.7	17.9	11.2	4.4	1.0	(25)
(26)			MCWB		-0.5	-0.2	1.5	10.0	15.2	17.9	19.9	18.4	14.3	9.4	3.7	0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		2.4	2.2	7.6	15.2	20.5	22.1	24.6	24.3	19.2	15.0	9.4	5.8	(27)
(28)			MCDB		2.8	3.0	11.4	21.7	27.3	28.5	30.3	30.6	23.2	18.0	10.6	6.2	(28)
(29)		2%	WB		1.0	1.1	4.5	13.2	18.8	20.9	22.9	21.9	17.4	12.6	6.9	3.5	(29)
(30)			MCDB		1.4	1.6	7.0	18.9	25.0	26.9	28.2	27.0	21.3	14.3	7.5	4.0	(30)
(31)		5%	WB		0.3	0.5	2.9	11.7	17.3	19.6	21.5	20.3	16.0	11.0	5.2	1.7	(31)
(32)			MCDB		0.7	1.0	4.6	16.6	22.6	24.8	26.4	24.9	18.9	12.7	5.8	2.2	(32)
(33)		10%	WB		-0.5	-0.1	1.8	10.2	15.8	18.4	20.3	18.9	14.6	9.6	3.7	0.5	(33)
(34)			MCDB		0.0	0.6	3.2	14.5	20.3	22.9	25.0	22.7	17.5	11.0	4.4	1.0	(34)
(35)	Mean Daily Temperature Range		MDBR		5.2	6.0	6.8	8.8	9.9	9.1	9.3	9.0	7.9	5.5	3.9	4.3	(35)
(36)		5% DB	MCDBR		3.8	4.5	8.3	11.9	12.5	11.9	11.9	12.1	10.9	8.5	5.0	3.7	(36)
(37)			MCWBR		3.8	4.0	5.6	6.7	6.5	5.6	5.6	6.0	6.2	5.9	4.6	3.5	(37)
(38)		5% WB	MCDBR		4.0	3.9	7.1	11.4	12.0	11.1	10.8	11.3	9.8	7.4	4.6	3.6	(38)
(39)			MCWBR		4.0	3.8	5.1	6.7	6.7	5.8	6.1	6.6	6.2	5.6	4.4	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.285	0.325	0.348	0.409	0.387	0.376	0.413	0.427	0.374	0.355	0.316	0.282	(40)
(41)		taud			2.091	2.047	2.072	2.128	2.324	2.379	2.296	2.237	2.364	2.330	2.216	2.087	(41)
(42)		Ebn at Noon			587	704	795	802	852	867	822	779	775	684	553	484	(42)
(43)		Edh at Noon			61	96	120	133	116	112	119	119	91	74	55	48	(43)
(44)	All-Sky Solar Radiation	RadAvg			0.57	1.46	3.01	4.01	5.40	5.47	5.49	4.36	2.70	1.24	0.48	0.32	(44)
(45)		RadStd			0.11	0.24	0.32	0.39	0.40	0.54	0.57	0.46	0.42	0.21	0.10	0.09	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	+0.73	N/A	N/A	N/A	N/A	N/A	N/A	-200	N/A	N/A	(47)

Nomenclature: See separate page